THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 10: **How Quantities Combine: Understanding Data** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 72	$\frac{24\times68+16\times78}{24+16}=72$
2	**(d)** 55	$\frac{18\times45+12\times70}{18+12}=55$
3	**(d)** 10	groups of equal size, so add the averages and divide by their number
4	**(a)** 7%	$\frac{2\times4+3\times9}{2+3}=7\%$
5	**(d)** 100	$\frac{200\cdot10+25y}{200+y}=15\Rightarrow y=100$
6	**(c)** 82	$\frac{90\times3+70\times2}{5}=82$
7	**(a)** 63	$\frac{15\times62+78}{16}=63$
8	**(c)** 10 litres	20×6% = 1.2 litres of solute; total volume = 30 litres
9	**(b)** mean depth of an irregularly shaped pit	Brāhmasphuṭasiddhānta (c. 628 CE)
10	**(d)** the proportions of the parts within each bar	every bar has the same total length (100%)

SECTION — B (Very Short Answer) — Answers 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{30\times70+20\times60}{50}=\frac{2100+1200}{50}=\mathbf{66}$ marks	1+1
12	Equal groups: $\frac{24+30}{2}=27$ minutes (2).	1+1
13	$\frac{200\times0.10+300\times0.20}{500}=\frac{20+60}{500}=\mathbf{16\%}$	1+1
14	$\frac{80\times2+60\times1+70\times1}{4}=\frac{290}{4}=\mathbf{72.5}$	1+1

SECTION — C (Short Answer) — Answers 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	The groups have different sizes (7 and 3), so the average must be weighted by group size (1). Correct: $\frac{70\times7+40\times3}{10}=\frac{490+120}{10}=61$ kg (2).	1+2
16	$\frac{40\times72+60\times80}{100}=\frac{2880+4800}{100}=76.8$ (2). It is nearer to 80 because Class B has more students (1).	2+1
17	Copper $=150\times0.6+250\times0.4=90+100=190$ kg (1½). Total $=400$ kg (½). Percentage $=\frac{190}{400}\times100=47.5\%$ (1).	1½+½+1
	OR: Let y litres be added: $\frac{10\times0.60+0.30y}{10+y}=0.50$ (1). $6+0.3y=5+0.5y\Rightarrow0.2y=1\Rightarrow y=5$ litres (2).	1+2
18	Weighted: $\frac{58\times3+66\times2+75\times5}{10}=\frac{174+132+375}{10}=68.1\%$ (2). Simple average $=\frac{58+66+75}{3}\approx66.33\%$, lower because the final exam, her best, has more weight (1).	2+1
	OR: Percentages: 80, 75, 90 (1). Weighted: $\frac{80\times2+75\times3+90\times5}{10}=\frac{160+225+450}{10}=83.5\%$ (2).	1+2

SECTION — D (Long Answer) — Answers 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Seniors: $168\times9=1512$ cm; juniors: $152\times6=912$ cm (2). (ii) $\frac{1512+912}{15}=\frac{2424}{15}=161.6$ cm (2). (iii) The groups are unequal in size, so each average must be weighted by its group size (1).	2+2+1
	OR: (i) Sum of first collection $=an$, of second $=bm$ (1½). Combined average $=\frac{\text{total sum}}{\text{total count}}=\frac{an+bm}{n+m}$ (1½). (ii) For $n=m$: $\frac{an+bn}{2n}=\frac{n(a+b)}{2n}=\frac{a+b}{2}$ (2).	3+2
20	(i) $\frac{100\times0.4+150\times0.2+0.6y}{250+y}=0.4$ (1½). (ii) $70+0.6y=0.4(250+y)=100+0.4y\Rightarrow0.2y=30\Rightarrow y=150$ kg (2). (iii) Silver $=40+30+90=160$ kg; total $=400$ kg; $\frac{160}{400}=0.4$ ✓ (1½).	1½+2+1½
	OR: (i) $\frac{3}{4}\times8=6\%$ (1). (ii) Spice $=0.8$ litre; new volume $=\frac{0.8}{0.06}=13.33$ litres, so add about 3.33 litres of water (3). (iii) About 13.33 litres (1).	1+3+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 10: **How Quantities Combine: Understanding Data** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 61	$\frac{20\times55+30\times65}{20+30}=61$
2	**(c)** 75	$\frac{25\times90+15\times50}{25+15}=75$
3	**(b)** 6%	$\frac{400\times5+100\times10}{400+100}=6\%$
4	**(b)** 16%	$\frac{150\times30+350\times10}{150+350}=16\%$
5	**(b)** 12	$\frac{6\cdot4+10y}{6+y}=8\Rightarrow y=12$
6	**(a)** 73.2	$\frac{66\times3+72\times2+78\times5}{10}=73.2$
7	**(a)** 42	$\frac{10\times40+62}{11}=42$
8	**(d)** 3 litres	9×20% = 1.8 litres of solute; total volume = 12 litres
9	**(a)** the simple average of the two group averages	the weights are equal
10	**(a)** Both A and R are true and R is the correct explanation of A	weights are the group sizes

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	With equal group sizes every group has the same weight, so the combined mean is the simple mean of the group means (1). With unequal sizes the larger group must count for more (1).	1+1
12	$3\times\frac{6}{100}=0.18$ litres of concentrate (2).	1+1
13	$\frac{9\times40+50}{10}=\frac{410}{10}=41$ (2).	1+1
14	Combined average $=\frac{an+bm}{n+m}$ (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Equal weights ⇒ simple mean (1): $\frac{\frac{1}{5}+\frac{3}{10}+\frac{1}{2}}{3}$ (1) $=\frac{\frac{2+3+5}{10}}{3}=\frac{1}{3}$ (1).	1+1+1
16	$\frac{6\times12+4\times14}{10}=\frac{72+56}{10}=12.8$ varṇa (3).	1+1+1
17	Guess: slightly above 40 (1). $\frac{20\times40+61}{21}=\frac{861}{21}=41$ km (2).	1+2
	OR: Spice $=1.2$ litres (1). $\frac{1.2}{12+y}=0.08\Rightarrow12+y=15$ (1), so $y=3$ litres (1).	1+1+1
18	$\frac{20\times150+30\times154+50\times158}{100}=\frac{3000+4620+7900}{100}=155.2$ cm (3).	1+1+1
	OR: $\frac{10\times30000+52000}{11}=\frac{352000}{11}=$ ₹32000 (3).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Bela, because the highest weight (5) is on skill where Bela scores 20 more, outweighing Aman's 20 extra in strength (weight 2) (1). (ii) Aman: $\frac{160+210+300}{10}=67$; Bela: $\frac{120+210+400}{10}=73$ (3). (iii) Bela (1).	1+3+1
	OR: (i) $\frac{46x+36(50-x)}{50}=40$ (1). (ii) $10x=200\Rightarrow x=20$ males; 30 females (2). (iii) $\frac{30\times36+42}{31}=\frac{1122}{31}\approx36.19$ kg (2).	1+2+2
20	(i) P: ₹4000; Q: ₹3000 (1). (ii) P: $\frac{1200}{4000}=30\%$; Q: $\frac{1500}{3000}=50\%$ (2). (iii) A stacked bar shows the totals (₹4000 vs ₹3000) with components; a 100% stacked bar shows the shares (30% vs 50%) (2).	1+2+2
	OR: (i) (a) can be inferred; (b) cannot (1½+1½). (ii) If A's total is 200 and B's 100: A's bread $=90$, B's $=30$ (b true); if A's total is 100 and B's 300: A's bread $=45$, B's $=90$ (b false) (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.